

6(a)(i)	nanometres / nm ;	1
6(a)(ii)	A ora throughout (presence of) carbohydrate / sugar, chains / residues, on, (glyco)proteins / (glyco)lipids ; A there are no sugar chains on the inner surface A (presence of) glycoproteins / glycolipids (on external surface) A (presence of) cell surface antigens / receptors (on external surface) A (presence of) glycocalyx	1
6(b)	R 1 cholesterol ; <i>one from</i> 2 maintains / regulates, fluidity of, membrane / phospholipid bilayer <i>or</i> at low temperatures, maintain / increase fluidity / prevents close packing A prevents hydrophobic 'tails' interacting at low temperatures <i>or</i> at high temperatures, stabilises the membrane / decreases fluidity ; 3 prevents passage (across membrane) of, hydrophilic / polar, substances ; S 4 phospholipid (monolayer) ; R phospholipid bilayer	4

Question	Answer	Marks
6(b)	<i>one from</i> 5 forms a bilayer ; 6 (bilayer is a) barrier to, water soluble molecules / polar molecules / ions ; ora not a barrier to lipid soluble molecules / allows lipid soluble molecules to cross membranes A non-polar / AW 7 (bilayer is a) non-polar barrier between cytoplasm and (aqueous) surroundings ; 8 hydrophilic 'head' forms hydrogen bonds with water ;	